When you go to production, you can implement everything in figure B.1 yourself, or there are providers that abstract this away for you, to varying degrees. However you deploy, the same concept is still happening underneath.

> **Serving static content from Django**
> When it comes to hosting architecture, there are plenty of choices, even the decision to serve static files from a web server isn't the only way. Django WhiteNoise is a third-party package that installs as middleware and serves static files directly from Django. It does this by bypassing all those extra steps I just mentioned. It isn't quite as fast as a web server, but it handles file changes and selective compression quite elegantly.

B.2 Server hosting vs. platform as a service

You can think of your hosting options as being on a spectrum. At one extreme (we'll call it the left end of the spectrum), you have the most control over your configuration and the most work to do. At the right end of the spectrum is *platform as a service* (PaaS), where you have very little work to do, but you more or less have to do it exactly the way the provider wants you to. The spectrum isn't quite evenly distributed; it's sort of chunky, and it can be divided into three pieces:

- Hardware
- Virtual private server
- Platform as a service

On the left side of the spectrum, you buy a server, plug it in, and connect it to the internet. Depending on your internet service provider, this may not even be allowed. You might plug it in at your place or rent a rack in a server room. Still very close to that left end is the virtual private server (VPS), where a hosting provider owns the hardware but gives you access to it. In the hardware case, and sometimes the VPS case, you are responsible for installing and maintaining the operating system, the web server, possibly Python, and the Django framework. The web server needs to be configured for your domain address and able to define base URL routes. The most common mappings have /static/ pointing to the static files on the web server, /media/ pointing to uploaded files, and all other URLs sent to Django through WSGI.

B.2.1 Virtual private servers

A VPS may or may not be shared. Shared servers are generally less expensive, but you run the risk of someone else on the server eating up all the CPU's cycles. Good hosting companies monitor this and keep on top of it, but it does still occasionally happen.

Even within the world of the VPS, there is a spectrum. Providers such as Linode and DigitalOcean (with their Droplets offering) start out with a Linux image, which you install on top of. They also offer pre-installed packages that include a web server, which means you have less to install yourself. A little further along the spectrum is a